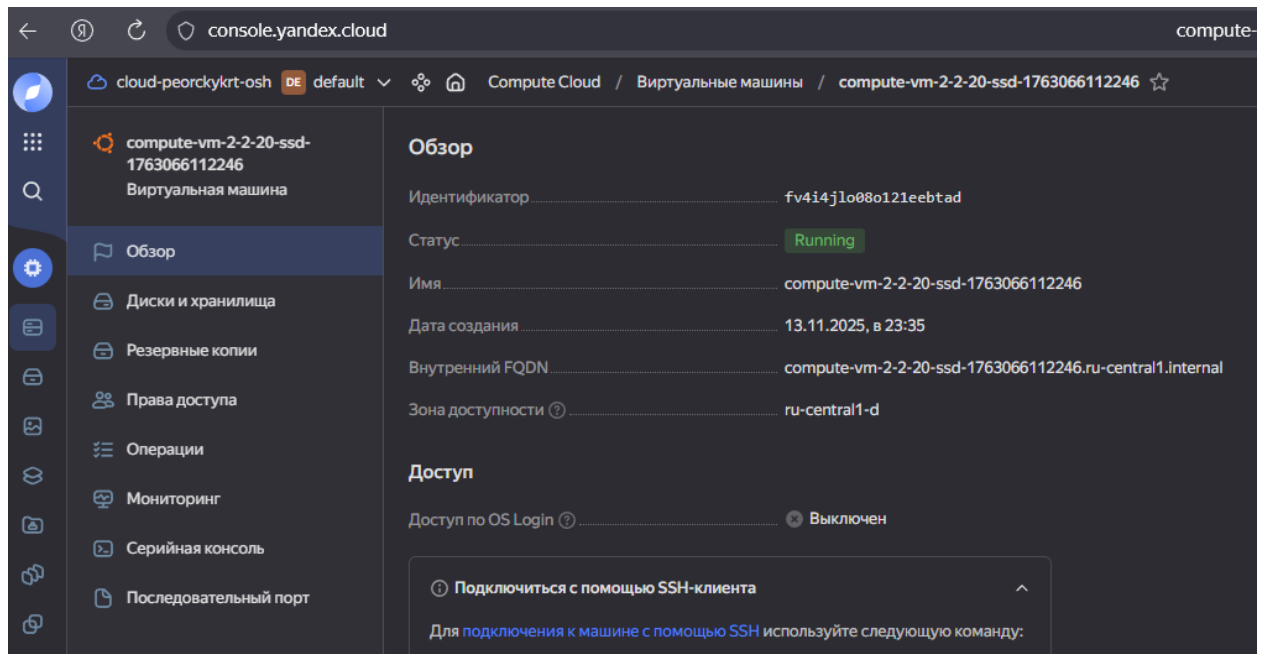
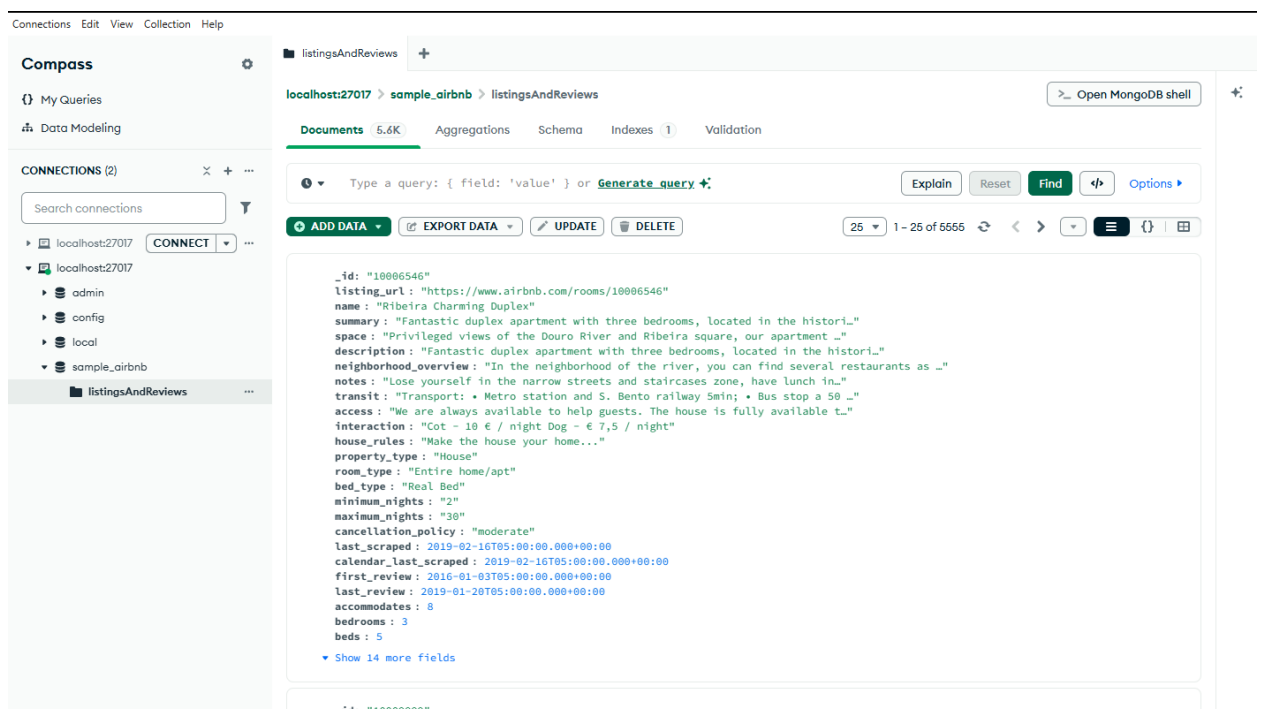


Шаг 1. Воспользуйтесь одним из вариантов ниже:

Создайте виртуальную машину в Yandex Cloud, установите на нее MongoDB.
Yandex Compute Cloud



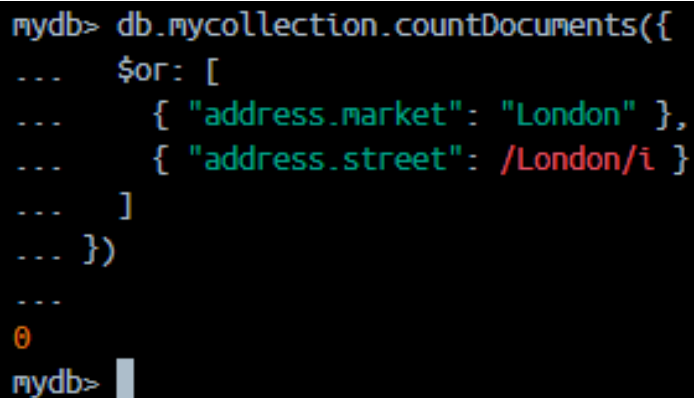
Шаг 2. Загрузите в MongoDB данные о листингах и отзывах AirBnB.



Шаг 3. Составьте запросы и найдите ответы на следующие вопросы (в домашней работе требуется как текст запроса, так и результат его исполнения). Рекомендуется использовать консольный клиент.

- Найти количество объектов, находящихся в Лондоне?

```
db.mycollection.countDocuments({
  $or: [
    { "address.market": "London" },
    { "address.street": /London/i }
  ]
})
```



```
mydb> db.mycollection.countDocuments({
...   $or: [
...     { "address.market": "London" },
...     { "address.street": /London/i }
...   ]
... })
...
0
mydb> 
```

Получается, что в Лондоне нет объектов.

Проверим есть ли Лондон вообще в списке уникальных значений address.market:

```
db.mycollection.distinct("address.market")
```



```
mydb> db.mycollection.distinct("address.market")
[
  '',
  'Barcelona',
  'Hong Kong',
  'Istanbul',
  'Kauai',
  'Maui',
  'Montreal',
  'New York',
  'Oahu',
  'Other (Domestic)',
  'Other (International)',
  'Porto',
  'Rio De Janeiro',
  'Sydney',
  'The Big Island'
]
mydb> 
```

Ответ: 0. В датасете нет объектов, находящихся в Лондоне.

- Найти количество объектов с ценой выше \$200.

```
db.mycollection.countDocuments({ price: { $gt: 200 } })
```

```
mydb> db.mycollection.countDocuments({ price: { $gt: 200 } })
---
1842
mydb> |
```

Ответ 1842 объекта.

- Найти объект с минимальной ценой.

```
db.mycollection.find().sort({ price: 1 }).limit(1)
```

Вывод:

```
[
  {
    _id: '14758068',
    listing_url: 'https://www.airbnb.com/rooms/14758068',
    name: 'Room on spacious apartment',
    summary: 'My space is close to restaurants and delicious bakeries. It is a big apartment, and there is a cat living there for now, so if you have animals, they are wellcome :) It is on quiet and safe place and has transports at the door. I can help with any need needed. Bills are included',
    space: '',
    description: "My space is close to restaurants and delicious bakeries. It is a big apartment, and there is a cat living there for now, so if you have animals, they are wellcome :) It is on quiet and safe place and has transports at the door. I can help with any need needed. Bills are included Guests can use the living room, kitchen and bathroom. They are shared and full equipped. I'll be happy to help with transports, interests, to discover Porto city. This place is quite unique. It has old streets and old houses. Stone houses, well preserved, narrow streets... you have to be curious and each walk will bring a discovery. There is a delicious bakery on the apartment street, at 7min walking. They open early, serve meals also and close late. They sell amazing and cheap cakes, there's wifi. Also there is a big supermarket next to the apartment, sells cheap products and has a big variety. There are parking places, for who have car. There are routes for bicycles, going to the center of Porto and to the",
    neighborhood_overview: "This place is quite unique. It has old streets and old houses. Stone houses, well preserved, narrow streets... you have to be curious and each walk will bring a discovery. There is a delicious bakery on the apartment street, at 7min walking. They open early, serve meals also and close late. They sell amazing and cheap cakes, there's wifi. Also there is a big supermarket next to the apartment, sells cheap products and has a big variety.",
    notes: 'As contas estão incluídas no preço do aluguer.',
    transit: 'There are parking places, for who have car. There are routes for bicycles, going to the center of Porto and to the center of Gondomar. And there are buses, right in front of the apartment.',
  }
]
```

```
access: 'Guests can use the living room, kitchen and bathroom. They are
shared and full equipped.',
interaction: "I'll be happy to help with transports, interests, to discover
Porto city.",
house_rules: 'A limpeza do apartamento deve ser feito pelos hóspedes.',
property_type: 'Apartment',
room_type: 'Private room',
bed_type: 'Real Bed',
minimum_nights: '30',
maximum_nights: '270',
cancellation_policy: 'flexible',
last_scraped: ISODate('2019-02-16T05:00:00.000Z'),
calendar_last_scraped: ISODate('2019-02-16T05:00:00.000Z'),
first_review: ISODate('2016-11-18T05:00:00.000Z'),
last_review: ISODate('2017-03-07T05:00:00.000Z'),
accommodates: 1,
bedrooms: 1,
beds: 1,
number_of_reviews: 2,
bathrooms: Decimal128('1.0'),
amenities: [
    'TV',
    'Wifi',
    'Kitchen',
    'Elevator',
    'Free street parking',
    'Washer',
    'Hangers',
    'Laptop friendly workspace',
    'translation missing: en.hosting_amenity_49',
    'translation missing: en.hosting_amenity_50',
    'Private entrance',
    'Hot water',
    'Microwave',
    'Coffee maker',
    'Refrigerator',
    'Dishwasher',
    'Dishes and silverware',
    'Cooking basics',
    'Oven',
    'Stove',
    'Patio or balcony',
    'Garden or backyard',
    'Long term stays allowed',
    'Wide hallway clearance',
    'Wide doorway',
    'Flat path to front door',
    'Well-lit path to entrance',
    'Disabled parking spot',
    'Wide doorway',
    'Wide clearance to bed',
```

```
    'Accessible-height bed',
    'Step-free access',
    'Wide doorway',
    'Accessible-height toilet',
    'Wide entryway',
    'Host greets you',
    'Handheld shower head'
  ],
  price: Decimal128('9.00'),
  security_deposit: Decimal128('100.00'),
  cleaning_fee: Decimal128('0.00'),
  extra_people: Decimal128('0.00'),
  guests_included: Decimal128('1'),
  images: {
    thumbnail_url: '',
    medium_url: '',
    picture_url: 'https://a0.muscache.com/im/pictures/1737a7fa-01a4-4dab-949d-
edee9d8a7712.jpg?aki_policy=large',
    xl_picture_url: ''
  },
  host: {
    host_id: '92014193',
    host_url: 'https://www.airbnb.com/users/show/92014193',
    host_name: 'Catarina',
    host_location: 'PT',
    host_about: '',
    host_response_time: 'within a few hours',
    host_thumbnail_url: 'https://a0.muscache.com/im/pictures/fdb1ce4f-5f10-
4d7d-b78e-30cda6c7f5c5.jpg?aki_policy=profile_small',
    host_picture_url: 'https://a0.muscache.com/im/pictures/fdb1ce4f-5f10-4d7d-
b78e-30cda6c7f5c5.jpg?aki_policy=profile_x_medium',
    host_neighbourhood: '',
    host_response_rate: 92,
    host_is_superhost: false,
    host_has_profile_pic: true,
    host_identity_verified: false,
    host_listings_count: 1,
    host_total_listings_count: 1,
    host_verifications: [ 'phone', 'reviews' ]
  },
  address: {
    street: 'Valbom, Porto, Portugal',
    suburb: '',
    government_area: 'Gondomar (São Cosme), Valbom e Jovim',
    market: 'Porto',
    country: 'Portugal',
    country_code: 'PT',
    location: {
      type: 'Point',
      coordinates: [ -8.557, 41.14586 ],
      is_location_exact: true
    }
  }
}
```

```

    }
  },
  availability: {
    availability_30: 0,
    availability_60: 0,
    availability_90: 26,
    availability_365: 301
  },
  review_scores: {
    review_scores_accuracy: 8,
    review_scores_cleanliness: 10,
    review_scores_checkin: 10,
    review_scores_communication: 10,
    review_scores_location: 8,
    review_scores_value: 10,
    review_scores_rating: 100
  },
  reviews: [
    {
      _id: '114415170',
      date: ISODate('2016-11-18T05:00:00.000Z'),
      listing_id: '14758068',
      reviewer_id: '97650684',
      reviewer_name: 'Marcel',
      comments: 'Ich durfte für einen Monat in der Wohnung von der Catarina
        übernachten und bin absolut zufrieden. \r\n' +
        'Seit dem ersten Schriftwechsel ist Catarina als eine sehr offene,
        sympatische und vertrauenswürdige Person aufgetreten, weswegen sie mir sowohl die
        Anreise erleichtert hat als auch dafür gesorgt hat, das ich mich in der neuen
        Umgebung wohlfühle. \r\n' +
        'Von der Ausstattung her ist alles so wie beschrieben. Einzige kleine
        Abänderung war eine Art Holzwand die zwischen dem eigenen kleinen Wohnbereich und
        dem Bereich für alle gezogen wurde. Dadurch wurde zwar der Aufenthaltsbereich
        verdunkelt aber ich hatte dadurch auch mehr Privatsspähre. \r\n' +
        'Die beiden Frauen auf die ich dort getroffen bin, sind auch mega nette
        Leute, mit denen ich mich vom ersten Tag an sehr gut verstanden habe.\r\n' +
        '\r\n' +
        'Alles in allem habe ich eine sehr schöne Zeit gehabt und kann die
        Wohnung definitiv weiterempfehlen.'
    },
    {
      _id: '135981722',
      date: ISODate('2017-03-07T05:00:00.000Z'),
      listing_id: '14758068',
      reviewer_id: '52845096',
      reviewer_name: 'Kai',
      comments: 'The host canceled this reservation 41 days before arrival.
        This is an automated posting.'
    }
  ]
}

```

```
]
```

Ответ: «Room on spacious apartment» с ценой \$9

- Найти город с максимальным количеством объектов.

```
db.mycollection.aggregate([
  {$group: {_id: "$address.market", count: {$sum: 1}} },
  {$sort: {count: -1}},
  {$limit: 1}
])
```

```
mydb> db.mycollection.aggregate([
...   { $group: { _id: "$address.market", count: { $sum: 1 } } },
...   { $sort: { count: -1 } },
...   { $limit: 1 }
... ])
...
[ { _id: 'Istanbul', count: 660 } ]
mydb> 
```

Ответ: Стамбул, 660 объектов.

- Найти объект с наибольшим количеством отзывов и вывести его название и количество отзывов

```
db.mycollection
.find({}, { name: 1, number_of_reviews: 1, _id: 0 })
.sort({ number_of_reviews: -1 })
.limit(1)
```

```
mydb> db.mycollection
...   .find({}, { name: 1, number_of_reviews: 1, _id: 0 })
...   .sort({ number_of_reviews: -1 })
...   .limit(1)
...
[ { name: '#Private Studio - Waikiki Dream', number_of_reviews: 533 } ]
mydb> 
```

Ответ: «#Private Studio - Waikiki Dream» с 533 отзывами.